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Comparison of Knowledge Transfer Methods for Bias Neutralization in Text

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Abstract. Bias in text can shape readers' perceptions and reinforce stereotypes, making automated text debiasing an important challenge in natural language processing. With the growing use of large language models (LLMs) for generating and editing content, the need for scalable neutrality-preserving methods continues to increase. Automated text debiasing with LLMs aims to reduce biased language while preserving semantic content. This paper presents a comparative study of multiple transfer learning paradigms for text neutralization using the Wikipedia Neutrality Corpus. We evaluate reinforcement learning approaches, Direct Preference Optimization (DPO) and Group Relative Policy Optimization (GRPO), with rewards provided by an LLM-as-a-Judge, alongside Chain-of-Thought (CoT) distillation, a supervised fine-tuned (SFT) model, and a pretrained baseline. Results show that GRPO achieves the strongest bias neutralization, followed by SFT, while DPO and CoT distillation fail to meaningfully improve neutrality despite strong lexical and semantic scores. Another striking observation is that traditional metrics exhibit weak correlation with neutrality performance, emphasizing the importance of task-specific evaluation and neutrality-aligned training signals for debiasing tasks.

Keywords. Text Debiasing, Supervised Fine-Tuning, Reinforcement Learning, Large Language Models, Bias Neutralization

1. Introduction

Bias in text poses a major challenge in natural language processing (NLP), as it can shape the reader's perception, reinforce stereotypes, and reduce objectivity. The growing prevalence of automatically generated and edited content further increases the need for scalable and reliable text debiasing methods. Large language models (LLMs) achieve strong performance across NLP tasks but inherently reflect biases present in their training data, often propagating subjective or imbalanced viewpoints. Mitigating such bias in LLM-generated or transformed text therefore remains both a technical and ethical challenge.

Reinforcement learning approaches such as Direct Preference Optimization (DPO) [1] and Group Relative Policy Optimization (GRPO) [2] enable models to learn from reward signals provided by a larger LLM acting as a judge [3]. In parallel, Chain-of-Thought (CoT) distillation [4] transfers reasoning processes from teacher to student models. These methods are particularly well suited for text neutralization, as they support nuanced linguistic alignment while remaining computationally efficient.

This work compares DPO, GRPO, and CoT distillation for biased text neutralization on the Wikipedia Neutrality Corpus (WNC) [5]. All methods are evaluated against a pretrained baseline and a supervised fine-tuned model using lexical metrics (BLEU, METEOR, ROUGE-L), semantic metrics (BERTScore and semantic similarity), and task-specific LLM-based evaluation. The source code is publicly available [6].

Relevance to Intelligent Environments. One natural setting for neutrality-preserving rewriting is online social platforms, which form an increasingly important communication layer of Intelligent Environments. Analogously to sentiment, politeness, and formality transfer proposed for toning down user-generated content and aligning posts with community expectations, bias neutralization can soften charged or subjective messages before they propagate, reducing miscommunication without heavy-handed moderation. A compact 1.7B-parameter neutralizer is well-suited to this role, as it fits the latency and compute budgets of interactive user-facing components.

2. Related Work

Bias in language models has been widely documented across dimensions such as gender [7], race [8], and other social attributes [9]. Prior work on bias mitigation includes adversarial training [10] and counterfactual data augmentation [11]. Supervised fine-tuning (SFT) remains a common approach for reducing bias in pretrained models [12], but it depends on curated labeled data and may fail to address biases not represented in the fine-tuning set.

Reinforcement Learning from Human Feedback (RLHF) is a prominent paradigm for aligning language models with human preferences [13], typically using Proximal Policy Optimization (PPO) [14]. However, RLHF can suffer from training instability, high computational cost, and challenges in defining reward functions for complex fairness objectives [15]. Group Relative Policy Optimization (GRPO) mitigates these issues by optimizing policies through within-group reward comparisons, removing the need for a separate value network and improving training stability. While GRPO has demonstrated effectiveness in domains such as mathematical reasoning, its application to fairness and bias mitigation remains largely unexplored.

Recent advances in reward modeling enable the use of language models as evaluators for alignment tasks [16, 17]. In this LLM-as-a-Judge paradigm, carefully designed prompts guide a large language model to assess outputs according to task-specific criteria. For bias neutralization, LLM-based judges offer a flexible mechanism for detecting subjective or imbalanced language and providing nuanced reward signals.

Knowledge distillation provides an alternative mechanism for transferring capabilities from larger teacher models to smaller student models. Chain-of-Thought (CoT) distillation [4] enables students to learn intermediate reasoning steps rather than only final outputs. This approach is particularly relevant for bias neutralization, as it facilitates the transfer of complex linguistic and ethical reasoning exhibited by larger models while maintaining output quality in more compact models.

3. Methodology

All knowledge transfer methods share the same base model, dataset and splits, teacher model, and parameter-efficient training scheme to enable a controlled comparison. Each method fine-tunes the Qwen3-1.7B base model using its native tokenizer, the WNC splits described below, and GPT-5-mini as the teacher model. All methods are trained with QLoRA (Quantized Low-Rank Adaptation) using 4-bit quantization and identical LoRA adapter configurations (rank, scaling, and target modules); per-method optimizer hyperparameters and sequence lengths are tuned individually and are reported in their respective subsections, since each paradigm has its own stability and memory profile.

We operationalize *neutrality* in line with the Wikipedia Neutral Point of View (NPOV) policy: the removal of subjective, evaluative, or loaded language while preserving the factual content, scope, and stance attributions of the source sentence. Under this definition, an ideal neutralization changes only the framing of an assertion, not what is being asserted.

3.1. Dataset: Wikipedia Neutrality Corpus

We use the Wikipedia Neutrality Corpus, a parallel dataset derived from Wikipedia revision histories in which biased or subjective sentences are edited to comply with Neutral Point of View (NPOV) guidelines. Each example consists of a biased source sentence paired with its neutral revision. Due to the size of the dataset, we use a deterministic slicing rather than the full corpus: the first 3,000 pairs of the canonical WNC ordering are used for training, the next 300 for validation, and the last 300 for held-out testing. The same fixed split is shared by all five methods and is reproducible from the released codebase; the impact of this single-seed choice is discussed in Section 5.4. This human-edited parallel structure makes the corpus well suited for evaluating bias neutralization while preserving semantic content.

3.2. Reinforcement Learning Methods

Reinforcement learning (RL) is commonly used to align language models with human preferences by optimizing model outputs according to task-specific reward signals. In text generation, the model acts as the policy, generated sequences correspond to actions, and rewards reflect output quality. We apply RL-based methods to bias neutralization using preference-based optimization and relative reward modeling.

3.2.1. Iterative DPO

Direct Preference Optimization (DPO) aligns language models using preference data without explicitly training a reward model, resulting in improved stability and efficiency compared to standard RLHF. In the iterative DPO setting, the model generates multiple candidate neutralizations per prompt using varied sampling temperatures (0.7, 1.0, 1.5). These candidates, along with the human-written neutral reference, are ranked by an LLM judge based on neutrality, semantic preservation, fluency, and minimal edits. Pairwise preferences are constructed from these rankings and used to optimize the model via the DPO objective. Training proceeds in batches of prompts, where generation, ranking, and optimization are performed iteratively using the model's current outputs. We use a learn-

ing rate of 1×10^{-6}, KL regularization parameter $\beta = 0.1$, and a maximum sequence length of 512 tokens. Mixed-precision training is employed depending on hardware. Unlike GRPO, DPO relies on binary preference comparisons rather than scalar rewards.

3.2.2. GRPO

Group Relative Policy Optimization (GRPO) is an RL method that optimizes model policies using rewards computed relative to a group of generated outputs, eliminating the need for a separate value network. For each prompt, the model generates multiple completions, and probability updates are driven by relative advantages within the group, with KL regularization applied to maintain proximity to a reference policy. We implement GRPO using the `GRPOTrainer` from the TRL library. The model generates three completions per prompt and is trained with a learning rate of 1×10^{-6} and clipping parameter 0.2. Rewards are provided by an LLM-as-a-Judge (GPT-5-mini), which evaluates outputs according to neutrality, preservation of meaning, fluency, and minimal edits (Table 1). Neutrality and meaning preservation receive the highest weighting.

Table 1. LLM-judge evaluation criteria and reward weights.

Criterion	Weight (%)
Neutrality	33.3
Meaning preservation	33.3
Fluency	16.7
Minimal edits	16.7

The judge assigns integer scores in the range $[-3, 3]$ for each criterion, which are combined into a weighted scalar reward and normalized to the $[0, 1]$ range. This formulation encourages neutral, semantically faithful outputs while minimizing unnecessary edits.

3.3. Chain-of-Thought Distillation

Chain-of-Thought (CoT) distillation transfers knowledge from a teacher model to a student by training on both the teacher's neutralized output and its accompanying reasoning. For each biased input, the teacher produces a neutralized sentence and an explanation of the applied changes, and the student is fine-tuned using supervised learning on this combined target. As with the other methods, the student is trained with QLoRA (4-bit quantization) using the shared LoRA adapter configuration (rank $r = 32$, scaling factor $\alpha = 16$). The student is trained for three epochs using a learning rate of 2×10^{-4}, an effective batch size of 32, and a linear learning rate schedule with warmup. Gradient checkpointing is applied, and the maximum sequence length is 2048 tokens. The model is optimized using a standard autoregressive language modeling objective.

3.4. Evaluation

Bias neutralization is evaluated using a hybrid framework that combines automatic metrics with LLM-based assessment. All models are evaluated on 300 samples from the Wikipedia Neutrality Corpus, with generated outputs and scores logged for analysis.

Automatic metrics compare model outputs against human-written neutral references using BLEU, METEOR, ROUGE (ROUGE-1, ROUGE-2, ROUGE-L), BERTScore, and sentence-level semantic similarity computed with all-MiniLM-L6-v2 embeddings. These metrics capture lexical overlap, semantic equivalence, and structural similarity.

To complement automatic evaluation, an LLM-as-a-Judge (Gemini 3 Flash) assesses outputs using the same criteria and weighting scheme employed for GRPO reward modeling. A different model is used for evaluation than for training rewards (GPT-5-mini) to avoid steering toward the training judge's preferences and to provide a more objective assessment. Results are tracked using MLflow, enabling structured comparison across models.

3.5. Implementation Details

Judge prompt. Both the GRPO reward judge and the final evaluation judge are prompted with the biased source, the model's neutralized output, and the human reference, and are asked to return integer scores in $[-3,3]$ for each of the four criteria together with a short justification. The four criteria are explicitly defined in the prompt (neutrality as removal of subjective or biased language, meaning preservation as factual fidelity, fluency and coherence, and edit minimality). The full prompt text is included verbatim in the released codebase [6].

Reward aggregation. Per-sample judge scores (s_n, s_m, s_f, s_e) are combined into the scalar reward $r = (s_n + s_m + 0.5\,s_f + 0.5\,s_e)/9 \in [-1,1]$, used directly as the GRPO signal; applied to the final evaluation judge's per-sample scores it yields the *Overall* column of Table 3.

Hardware. All training runs are performed on a single GPU with bf16 support (compute capability ≥ 8.0, e.g., NVIDIA A100/H100); QLoRA 4-bit quantization keeps the working set within a single accelerator for all five methods.

4. Results

Table 2. Automatic evaluation metrics across Qwen3-1.7B models (300 test samples). MET. = METEOR; RG-*n* = ROUGE-*n*; BERT = BERTScore F1; Sem. = semantic similarity. Best per metric in **bold**.

Model	BLEU	MET.	RG-1	RG-2	RG-L	BERT	Sem.
Base	17.50	71.02	68.75	57.51	67.09	80.96	81.72
SFT	56.86	73.02	73.37	63.53	72.31	83.85	83.65
Iter. DPO	**65.15**	**88.69**	**86.71**	**78.78**	**85.97**	**91.92**	**93.62**
GRPO	28.72	70.20	70.75	55.15	67.90	84.38	88.28
CoT	30.87	72.60	72.43	56.48	69.64	85.35	90.61

4.1. Base Model

The pretrained Qwen3-1.7B model serves as a reference point. As shown in Table 2, lexical similarity with reference texts is low, while semantic metrics indicate moderate meaning preservation. LLM-as-a-Judge evaluation yields an overall score of 0.63, driven by strong meaning preservation and fluency but near-zero neutrality (0.01), indicating that the base model produces coherent outputs but fails to address bias.

Table 3. LLM-as-a-Judge mean scores across Qwen3-1.7B variants (300 test samples). Criteria scored on $[-3, 3]$; Overall on $[0, 1]$.

Metric	Base	SFT	Iter. DPO	GRPO	CoT
Neutrality	0.01	0.67	0.20	**1.14**	0.29
Meaning Pres.	2.79	2.55	**2.88**	2.70	2.87
Fluency	2.90	2.88	2.88	**2.96**	2.82
Minimal Edits	2.75	2.69	2.72	2.21	**2.76**
Overall	0.63	0.67	0.65	**0.71**	0.66

4.2. Supervised Fine-Tuning

Supervised fine-tuning on the WNC dataset improves both lexical and semantic alignment with reference texts compared to the base model. LLM-based evaluation yields an overall score of 0.67, the second highest among all methods, with notably improved neutrality (0.67). However, this comes with slightly lower meaning preservation, suggesting a trade-off between bias removal and content fidelity.

4.3. Iterative DPO

Iterative DPO achieves the highest scores on all lexical and semantic metrics, indicating strong preservation of reference wording and meaning. However, LLM-as-a-Judge evaluation reveals weak neutrality (0.20) with a median of 0.00, despite high fluency and meaning preservation. The overall judge score of 0.65 is the second lowest among fine-tuned models.

4.4. GRPO

GRPO achieves the strongest bias neutralization among all methods. While lexical overlap with reference texts is moderate, semantic metrics confirm robust meaning preservation. The LLM-as-a-Judge assigns the highest overall score (0.71), with a neutrality score of 1.14 that substantially exceeds all other approaches. GRPO also achieves the highest fluency (2.96) with the lowest variance (0.25), though edit minimality is lower (2.21), suggesting a tendency toward more extensive rewrites.

4.5. Chain-of-Thought Distillation

Chain-of-Thought distillation achieves strong semantic preservation (2.87) and high edit minimality (2.76), performing comparably to Iterative DPO on these dimensions. However, the LLM-as-a-Judge yields a moderate overall score of 0.66, with a low neutrality score (0.29) and a median of 0.00, performing similarly to the base model and DPO on the neutrality criterion.

4.6. Qualitative Examples

Table 4 illustrates the per-method behavior on two representative test sentences. In both cases the bias is concentrated in a single loaded modifier (*brilliant breakthrough*; *obvious parody*), and only GRPO consistently rewrites the sentence to remove it; SFT also suc-

ceeds on the second example. Iterative DPO, CoT distillation, and the base model leave the biased phrasing essentially intact, which is consistent with their high edit minimality scores and low neutrality scores in Table 3.

Table 4. Qualitative outputs on two test sentences. Per-sample neutrality score (in $[-3, 3]$) from the evaluation judge in parentheses.

Example 1. Biased: *berners-lee's brilliant breakthrough was to marry hypertext to the internet.* Reference: *berners-lee's breakthrough was to marry hypertext to the internet.*	
Base (-2)	berners-lee's brilliant breakthrough was to marry hypertext to the internet.
SFT (-3)	berners-lee's brilliant breakthrough was to marry hypertext to the internet.
Iter. DPO (-3)	berners-lee's brilliant breakthrough was to marry hypertext to the internet.
GRPO ($+2$)	Berners-lee's groundbreaking contribution was to link hypertext with the internet.
CoT (-2)	berners-lee's brilliant breakthrough was to marry hypertext to the internet.
Example 2. Biased: *the death egg is an obvious parody of the death star space station in the star wars series.* Reference: *the death egg is a parody of the death star space station in the star wars series.*	
Base (-2)	The death egg is an obvious parody of the death star space station in the star wars series.
SFT ($+3$)	the death egg is a parody of the death star space station in the star wars series.
Iter. DPO (-2)	The Death Egg is an obvious parody of the Death Star space station in the Star Wars series.
GRPO ($+3$)	The Death Egg is a parody of the Death Star space station in the Star Wars series.
CoT (-1)	the death egg is an obvious parody of the death star space station in the star wars series

5. Discussion

5.1. Comparison of Automatic Metrics

Automatic metrics reveal a clear split between lexical overlap and semantic preservation. As shown in Table 2, Iterative DPO leads on all surface-level metrics, while GRPO and CoT obtain lower BLEU but comparable METEOR, suggesting greater lexical diversity with semantically valid outputs. Notably, GRPO and CoT substantially outperform SFT on semantic similarity and BERTScore, indicating that reinforcement learning and reasoning-based distillation promote deeper semantic understanding even when lexical overlap is lower. Crucially, the disconnect between automatic metrics and LLM-based evaluation is stark: Iterative DPO leads on all lexical and semantic metrics but ranks among the lowest on neutrality. This confirms that surface similarity to reference text does not imply successful bias removal.

5.2. LLM-as-a-Judge Evaluation

While overall scores are relatively close across methods (0.63–0.71), neutrality performance varies dramatically (Figure 1). SFT emerges as the second-best method for neutrality despite its simplicity, outperforming both DPO and CoT. Iterative DPO and CoT achieve strong meaning preservation and edit minimality but fail to address bias effectively, with median neutrality scores of 0.00. This suggests that optimizing for preference alignment or transferring reasoning chains does not inherently lead to bias removal without explicit neutrality-focused training signals.

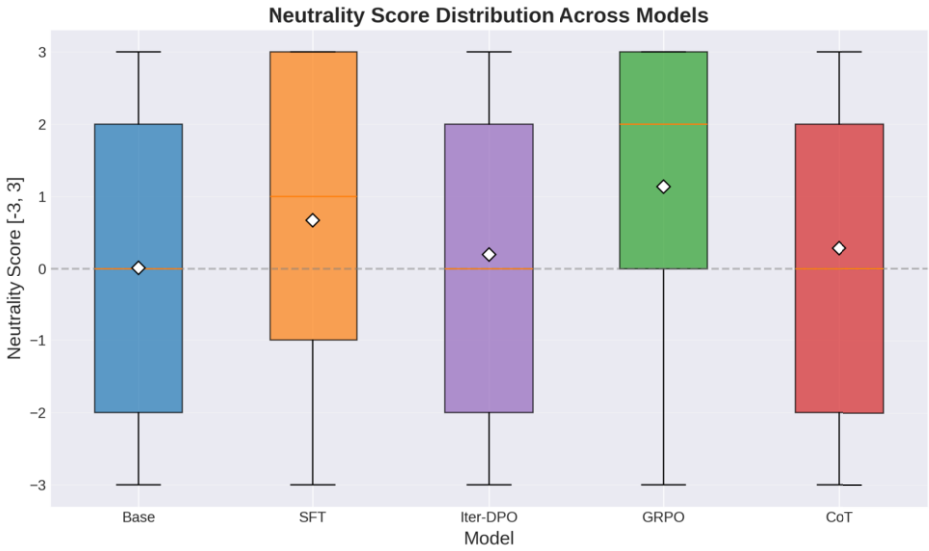


Figure 1. Distribution of neutrality scores across models. GRPO is the only method with a consistently positive neutrality distribution.

GRPO's advantage stems from its direct optimization of neutrality through reward signals. As shown in Table 3, this comes at the cost of lower edit minimality, as GRPO produces more extensive rewrites. In contrast, methods with high edit minimality (DPO, CoT, base) tend to preserve the original biased phrasing, resulting in low neutrality. This inverse relationship between edit minimality and neutrality suggests that effective debiasing inherently requires more substantial textual changes.

5.3. Correlation between Automatic Metrics and Neutrality

To directly quantify the disconnect between surface-level metrics and neutrality, we compute pooled Pearson (r) and Spearman (ρ) correlations between each per-sentence automatic metric and the per-sentence neutrality score assigned by the LLM judge, across all five methods ($n = 1500$). Confidence intervals are obtained via the Fisher z-transform. Results are shown in Table 5.

BLEU, METEOR, ROUGE-1, ROUGE-2, and ROUGE-L all exhibit small but statistically significant *negative* correlations with neutrality (Pearson r between -0.20 and -0.11). In other words, sentences with higher lexical overlap with the human-written neutral reference tend to receive *lower* neutrality scores, not higher. BERTScore F1 and semantic similarity are statistically indistinguishable from zero: both 95% confidence intervals cross zero for Pearson and Spearman alike. Taken together, these coefficients substantiate the observation associated with Table 2 that standard automatic metrics are poor proxies for bias neutralization. Surface similarity to a neutral reference is not a reliable signal of successful debiasing, and embedding-based metrics capture no useful information about neutrality on this task.

Table 5. Pooled Pearson (r) and Spearman (ρ) correlations between per-sentence automatic metrics and the LLM-judge neutrality score across all five methods ($n = 1500$). 95% confidence intervals computed via the Fisher z-transform. **Bold** entries are significant at $p < 0.05$.

Metric	Pearson r	95% CI	Spearman ρ	95% CI
BLEU	**-0.151**	[-0.20, -0.10]	**-0.140**	[-0.19, -0.09]
METEOR	**-0.202**	[-0.25, -0.15]	**-0.200**	[-0.25, -0.15]
ROUGE-1	**-0.106**	[-0.16, -0.06]	**-0.078**	[-0.13, -0.03]
ROUGE-2	**-0.133**	[-0.18, -0.08]	**-0.108**	[-0.16, -0.06]
ROUGE-L	**-0.112**	[-0.16, -0.06]	**-0.079**	[-0.13, -0.03]
BERTScore F1	-0.030	[-0.08, 0.02]	0.004	[-0.05, 0.05]
Sem. Sim.	-0.039	[-0.09, 0.01]	-0.016	[-0.07, 0.03]

5.4. Limitations

Several limitations should be considered when interpreting these results. **(i) No human calibration of the judge.** Neutrality is subjective and the LLM-as-a-Judge may carry its own biases; we mitigate the most direct form of this bias by using a different judge for evaluation (Gemini 3 Flash) than for training rewards (GPT-5-mini), but provide no human-rater validation or inter-rater agreement, so the ranking is best read as the ranking under this specific judge-based operationalization of neutrality. **(ii) Train–evaluation criterion overlap.** GRPO is trained on a reward computed from the same four criteria and weighting scheme later used for evaluation; despite the judge-model swap, the training signal is structurally aligned with the evaluation rubric, which may inflate GRPO's apparent neutrality margin. **(iii) Single split, single seed.** All methods are trained and evaluated on a single deterministic 3,000/300/300 slicing of WNC; we therefore do not report mean $\pm$ standard-deviation across seeds or subset draws. **(iv) Single dataset and model family.** All experiments use the Wikipedia Neutrality Corpus and a single backbone (Qwen3-1.7B); the ranking may not transfer to other domains, bias types, or architectures. Broader evaluation across human annotators, datasets, judges, seeds, and model families is needed to confirm the generality of these findings.

6. Conclusion

This study compared five knowledge transfer methods for bias neutralization on the Wikipedia Neutrality Corpus. GRPO emerges as the most effective approach, achieving the highest overall score and the strongest neutrality, while SFT ranks second, demonstrating that direct supervision on parallel neutral data provides a reliable improvement. Iterative DPO and CoT distillation perform poorly on neutrality despite strong lexical and semantic scores, confirming that surface-level similarity metrics do not capture bias removal. The findings demonstrate that effective debiasing requires training signals explicitly aligned with neutrality objectives. Without such signals, methods default to preserving surface form rather than addressing bias. From an Intelligent Environments perspective, a compact neutralizer of this kind could serve as a lightweight tone-down layer for user-generated text in online social platforms, complementing existing sentiment- and politeness-style-transfer approaches without the latency overhead of larger models. Future work should explore combining GRPO's neutrality-focused optimization with the

semantic fidelity of distillation-based methods, and validate the resulting systems with human raters in deployment-like social-platform settings.

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